R01 RESUBMISSION · ANTIMICROBIAL RESISTANCE

Programmable phages to disarm antibiotic-resistant infections

A three-aim program to turn bacteriophages into a tunable last-line therapy.

Significance: resistant infections are rising, and the antibiotic pipeline is empty.

Carbapenem-resistant *Klebsiella* now causes infections with no reliable cure, and no new antibiotic class has reached the clinic in over a decade. We propose engineered bacteriophages — viruses that hunt bacteria — as a programmable therapy that resistance cannot easily outrun.

The gap: natural phages work, but we cannot control them well enough for the clinic.

Compassionate-use cases prove phages can clear resistant infections. But natural phages have narrow, unpredictable host ranges, and bacteria evolve resistance to them within days. Phage therapy stalls not on biology but on *controllability* — and that is an engineering problem.

- Host range is narrow and hard to predict.
- Bacterial resistance to a single phage emerges in days.

What this program must establish, across three aims.

01

Retargetable host range

Aim 1 — engineer phage tail fibers to bind resistant *Klebsiella* strains on demand.

02

Resistance-proof design

Aim 2 — build phage cocktails that require three simultaneous mutations to escape.

03

Safe in vivo clearance

Aim 3 — demonstrate efficacy and safety in a murine bacteremia model.

Aim 1: a modular tail-fiber platform that retargets phages to new strains.

We will build a library of swappable receptor-binding domains so a single phage chassis can be reprogrammed to any clinical isolate within 48 hours. Our preliminary data already shows retargeting against three of five test strains.

- Modular chassis plus a domain library, swapped by cloning.
- Preliminary: 3 of 5 strains retargeted successfully.

Aim 2: cocktails engineered so escape requires three independent mutations.

Single-phage therapy fails because one mutation defeats it. We will design three-phage cocktails that bind three independent, essential receptors — forcing the bacterium to mutate all three at once to escape. The math makes that vanishingly unlikely.

- Three orthogonal receptors per cocktail.
- Predicted escape frequency below one in a trillion.

Aim 3: efficacy and safety in a murine bacteremia model.

We will infect mice with a lethal carbapenem-resistant strain and treat with the engineered cocktail, measuring survival, bacterial clearance, and immune response. Endotoxin release and off-target effects are the safety endpoints we will watch most closely.

- Primary endpoint: 7-day survival versus untreated controls.
- Safety: endotoxin, cytokine, and off-target screens.

How the three aims unfold across the funding period.

AIM	YEARS 1–2 BUILD	YEARS 2–4 TEST	YEARS 4–5 TRANSLATE
Aim 1 — Retargeting	Shipped: Domain library	In flight: Full strain panel	Planned: Rapid-response protocol
Aim 2 — Cocktails	In flight: Three-receptor design	Planned: Escape assays	Planned: Stability formulation
Aim 3 — In vivo	Planned: Model setup	Planned: Efficacy study	Planned: IND-enabling tox

Shipped · In flight · Planned

Rigor and feasibility: why this team can deliver.

Our lab built the modular chassis; our co-investigator runs the regional phage biobank with 400 characterized isolates; our murine model is established and IACUC-approved. Every aim rests on a capability we already demonstrate in preliminary data.

- Biobank of 400 isolates de-risks Aim 1.
- Established, approved animal model de-risks Aim 3.

What a funded program delivers in five years.

<48 h

Retarget time

ANY CLINICAL ISOLATE AIM 1

<1e-12

Escape frequency

THREE-RECEPTOR COCKTAIL AIM 2

80%

Murine survival

VS 0% UNTREATED AIM 3

Innovation: we move phage therapy from artisanal to programmable.

Today each phage treatment is hand-matched to a patient over weeks. This program delivers a platform that retargets in days and resists escape by design — the difference between a craft and a manufacturable medicine.

- From weeks of hand-matching to a 48-hour protocol.
- A platform, not a one-off compassionate-use case.

PI · DR. SAMUEL ADEYEMI · CENTER FOR ANTIMICROBIAL ENGINEERING

When antibiotics run out, a programmable predator is the next line.